

Application of Robotic Process Automation in collection of students satisfaction at University

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Abstract - Currently, it is important for universities to understand, analyze, evaluate and improve student satisfaction. In order to have a basis for improving student satisfaction, universities need to collect data on students' emotions. However, due to the large number of students who need to be surveyed, the number of questions in each survey is large, so the scope of the survey usually stops at assessing the student's satisfaction level related to the lecturer and the course. Survey questions may not be tied to students' emotions at some key moments. Besides, this data collection is usually done directly in the classroom or sending online survey forms to students. This leads to an increased workload of the staff in charge of collecting the surveying students, while there may still be a large number of students who do not take the surveys due to absence or disinterest. So "How can we 'encourage' students to answer the questions? How does the staff in charge simplify the collection of feedback? How can university leaders regularly capture student satisfaction in real time?". This study was conducted to address the above challenges based on the application of Robotics Process Automation (RPA) technology in defining the journey of collecting students' feedback, automating sending and receiving and analyzing student satisfaction data. The application of the above technology is expected to bring interesting experiences for students; help university leaders assess whether the products and services they provide to students are good or not, are there any adjustments needed? In addition, the study will show how to experimentally implement the problem of collecting student satisfaction feedback based on the akaBot platform.

Keywords - Robotic Process Automation, RPA, Student satisfaction

I. INTRODUCTION

In general, the quality of education is the top factor that universities aim to bring the best benefits to students. From there, it helps to form prestige and competitive advantages for universities in attracting students, as well as receiving positive reviews from employers and society. To achieve the above goal, besides focusing on improving training content and programs; the enhancement of teaching quality of lecturers, improvement of services related to students, creating a favorable learning environment for students, bringing satisfaction to students - have been done by the universities and concretized through vision, mission and strategy statements. Therefore, one of the most important tasks to perform is how to understand, analyze, evaluate and improve student satisfaction. The above work will help identify the gap between students'

expectations about the "training" service and their perception of the results when using the "training" service that the higher education institution provides. In order to do this, universities often conduct activities to collect data on students' opinions, views and feelings through survey questionnaires at the end of courses. In the case of Vietnam, according to statistical data from the Ministry of Education and Training, the current training scale is 1.526.111 students, at 237 universities, each university has an average of more than 6.400 students [1]. So, the difficulty that universities often face is the need to conduct thousands of surveys (*equivalent to the number of classes opened for students*), thousands of hours to input, synthesize and analyze data related to students' opinions. On the other hand, the survey forms often have a large number of questions, the content of some questions may not be clear, the answer options may be quite qualitative, etc. Therefore, students may lose focus in patiently answering each question correctly.

So how can university leaders regularly grasp, synthesize and adjust policies based on the evolution of students' views; staff assigned the task of collecting comments from students can reduce the workload; lecturers who can see the point of students' feelings can do it often and quickly; students can answer more easily, conveniently and accurately? The above expectations can be effectively addressed through analyzing the process of gathering students' opinions, optimizing the process based on finding out which manual tasks can be converted to automated implementation based on RPA technology.

In fact, RPA is a topic of particular interest in the Business Process Management (BPM) community [2]. RPA is a platform for creating software robots (bots) that automate well-defined, regular and repetitive manual tasks without the need for human cognitive action ([3], [4], [5]). Software robots created by RPA technology can perform tasks in a single process or across multiple processes in different time intervals, with faster and more stable speeds without rest. RPA offers many benefits in terms of cost and efficiency, allowing users to focus on tasks that require thinking and creativity by eliminating boring repetitive tasks ([6], [7]). RPA is suitable for solving "non-invasive" automation approaches as opposed to traditional automation solutions - which do not require major changes to existing systems [8]. Currently, organizations can easily build, deploy and manage software robots based on several RPA platforms like UiPath, BluePrism, Automation Anywhere, etc.

This study focuses on answering two research questions related to the above mentioned content:

- (1) How to apply RPA technology to automate sending surveys to students?
- (2) What is the effect of applying RPA technology in practice?

II. METHODOLOGY

The objective proposes an information system to develop a solution to realize the creation, automatic sending, collection and visualization of feedback related to student satisfaction; the research team chooses Design Scientific Research (DSR) method in developing the information system ([9], [10]). This approach is proven to be effective in designing information technology (IT) solutions that address real-world problems ([9], [10], [11]). To concretize the implementation, the DSR process has been applied to solve the set goal [12]. The tasks performed and the corresponding outputs in the DSR process are shown in Fig. 1.

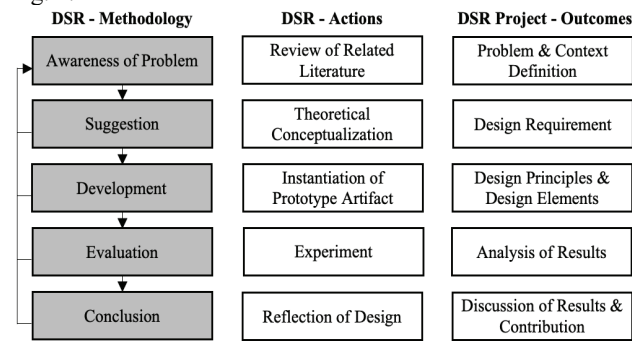


Fig. 1. General DSR approach. Adapted from Kuechler and Vaishnavi (2008)

Step 1: Awareness of Problem

As analyzed in the introduction, the problem mentioned in this study is related to the target groups of university leaders,

officer staff, lecturers, students. The goal is how to be able to record the level of student satisfaction. In order to discover the problem that needs to be solved, the research team conducted a survey of the implementation situation at a number of universities in the North of Vietnam in the period from 2013 to 2023. The research team has conducted a synthesis of the forms of collecting student opinions according to the group of universities with the level of IT application. The survey results can be presented in Table I.

Through the survey results, it can be seen that the data collection of students' opinions is done with quite different results, from not doing (*University group 1*); do it manually (*University group 2*); performed well but took a lot of work to synthesize (*University group 3*) and performed quite well (*University group 4*). It is often difficult for university leaders to grasp the progress of students' opinions (*when waiting for reports from functional units or coming to each semester*); office staff (*in University group 2, University group 3*) are quite difficult to implement; lecturers have not received the attention and recommendations in collecting and using the results of students' feedback (*self-implementation, optional*). In addition, the implementation of the activity of collecting students' opinions is still fragmentary in initializing, collecting, evaluating results; manual in sending feedback; adding more complex survey features if you want to integrate in the university's existing information systems.

Step 2: Suggestion

With the discovery of some existing problems in step 1, the research team expects that there should be a mechanism for university leaders to grasp the evolution of students' satisfaction with the university's activities in real-time (*can be displayed on a dashboard of a web-based system*). The creation, collection and visualization of data should also be done in a centralized, unified place (*on a web-based system*). Sending feedback to students can use RPA technology to automatically do; the software robot will access the web-based system to get the data and send an email asking each student for feedback.

TABLE I. SUMMARY OF FORMS OF COLLECTING STUDENTS' OPINIONS AT SOME UNIVERSITIES IN VIETNAM

University group	University leaders		Office staff		Lecturers	
	How to capture students' opinions?	Frequency	How to collect students' opinions?	Frequency	How to collect students' opinions?	Frequency
University group I (very limited level of IT application)	Based on the report of the functional department	Regulated (term, academic year)	Do not accept the opinions of students	Do not perform	The lecturer decides on the form of collecting students' opinions (self-implementation, optional)	Instructors decide how often to collect students' opinions (self-executing, optional)
University group II (limited IT application)			Deploy student opinion polls via questionnaires directly in the Lecture Hall (students may not be required to take votes if absent)	End of term		
University group III (normal level of IT application)			Implement taking feedback from students for each course via Google Form (students may not be required to take votes)	End of term		
University group IV (good level of IT application)	Based on departmental reports or data on the system	Regulated (term, academic year) or real time	The university deploys to collect students' opinions via the Form on the Student Portal (if students do not evaluate, they will not be able to register in the new semester).	End of term		

Step 3: Development

The university needs to develop two systems:

- **Web-based system:** has the feature of decentralization for user groups (*university leaders, staff, lecturers*), can create student survey forms, collect survey data and data visualization.
- **Software robot (developed on RPA technology):** capable of automatically accessing the web-based system, reading if there is a survey that needs to be done, getting survey data and sending email to the person in need of the survey.

The above two systems need to be deployed from system design activities (*interfaces, databases, etc.*), to programming, testing and practical implementation. It will help to thoroughly solve the problem posed at the beginning.

Step 4: Evaluation

Each university can conduct experiments to survey students' opinions periodically or irregularly; can survey according to job requirements for departments; or simply help lecturers get a tool to record the level of understanding, the level of interest,

the level of satisfaction in each lesson, etc. From there, synthesize and visualize the data according to each survey; or aggregate all surveys.

The implementation of system development, also helps the university to check the errors in the implementation process, the appropriate level of content, frequency, etc. of the surveys. Thereby, there are appropriate adjustments.

Step 5: Conclusion

Thanks to the data that is always updated, the university can make appropriate adjustments to help students feel cared about, improve satisfaction, increase sympathy and understanding.

III. RESULTS

A. Proposing the process of surveying students' opinions with RPA support

After identifying the problem and considering the appropriate implementation solution, "How should the creation, sending, collection and visualization of data on survey content related to student satisfaction be done with RPA support?", the research team proposed a process of surveying students' opinions as shown in Fig. 2.

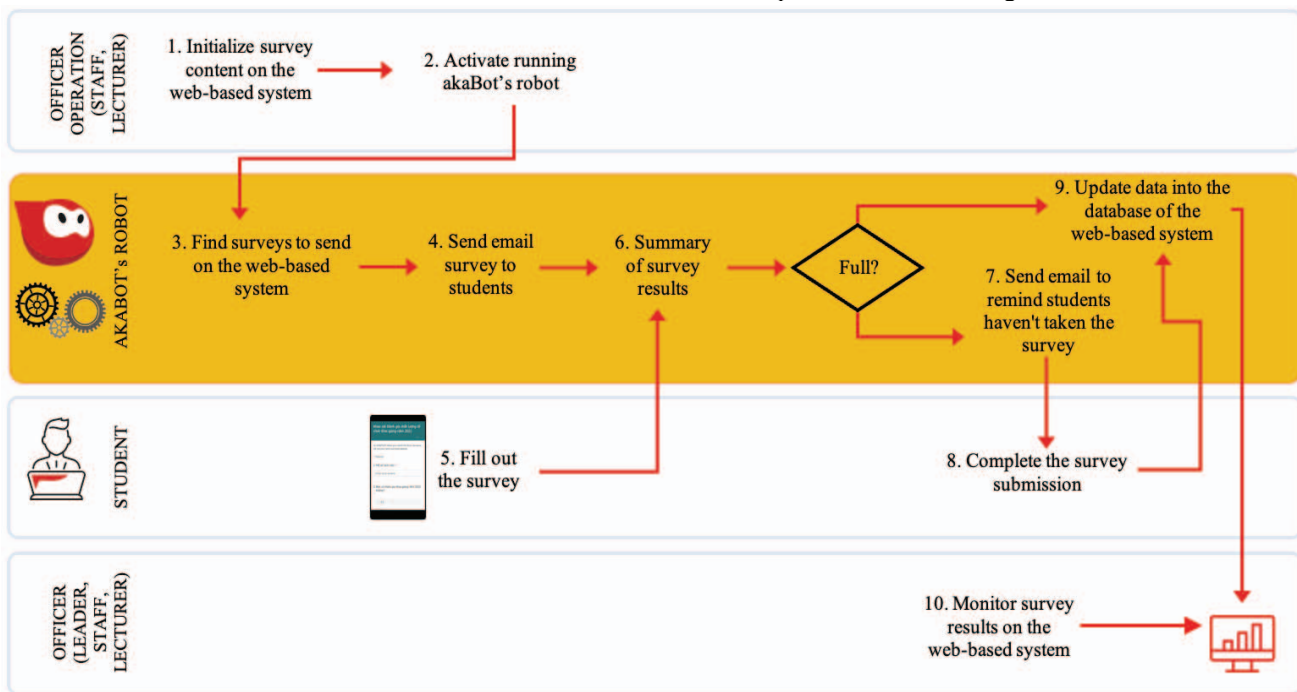


Fig. 2. Proposed process for surveying students' opinions with RPA support

The proposed process consists of 10 steps including the participation of:

- **Leaders, staff, lecturers:** create content and evaluate survey results.
- **Software robot:** built on RPA technology (*named AKABOT's ROBOT, built on akaBot solution - one of the 30 best RPA platforms in the world, a product of a*

famous Vietnamese technology company) to automatically send surveys to students.

- **Students:** conduct surveys.

The main tasks in the process of surveying students' opinions proposed by the research team are detailed in Table II below.

TABLE II. SPECIFICATION OF STUDENT SURVEY PROCESS

Step	Execution object	Implementation content	Note
01	Officer operation (Staff, Lecturer)	Initialize survey content on the web-based system	
02	Officer operation (Staff, Lecturer)	Activate running akaBot's robot	
03	akaBot's robot	Find surveys to send on the web-based system	
04	akaBot's robot	Send email survey to students	
05	Student	Fill out the survey	Students only need to select the "icon" that matches their point of view
06	akaBot's robot	Summary of survey results	- If not complete, proceed to step 07 - If complete, perform step 09
07	akaBot's robot	Send email to remind students haven't taken the survey	
08	Student	Complete the survey submission	
09	akaBot's robot	Update data into the database of the web-based system	
10	Officer (Leader, Staff, Lecturer)	Monitor survey results on the web-based system	Collected data is visualized by each survey or all surveys

With the proposed process, universities can adjust to suit their specifics such as increasing/decreasing the time students spend doing surveys or changing the platform for building software robots to automate some tasks related to sending surveys to students, etc.

B. Developing a web-based system to initiate, synthesize feedback and visualize data

Aim to establish a unified communication channel for leaders, staff, and lecturers with the system. The research team proposes to develop a web-based system to perform 3 main functions: initializing survey content; synthesizing feedback from students and visualizing the data obtained in real time.

- **Initialize survey content:** the system allows operators (staff, lecturers) to select students to receive feedback, survey title, survey start/end time, time check which students have not taken the survey to send back, survey content. As for the survey content, the author team automatically integrates the APIs in the "icons" to quickly identify and synthesize students' opinions related to the survey.
- **Synthesize feedback from students:** when students click/touch on "icons" related to students' opinions about the survey; the API is automatically enabled to record students' responses. The web-based system will immediately summarize the feedback of students in real time.
- **Data visualization:** with feedback data obtained from students, the web-based system automatically updates the data chart related to each survey or aggregates all

surveys. As a result, leaders, officials and lecturers can grasp the students' views; then make the appropriate adjustments.

TABLE III. SPECIFICATION OF SOME MAIN FEATURES OF THE WEB-BASED SYSTEM

Feature	Recommended interface
Initialize survey content	
Summarize feedback from students	
Data visualization	

C. Applying RPA platform in automatically sending survey content to students

For this study, the authors chose to use the akaBot solution, one of the top 30 RPA platforms along with leading solutions like UiPath, BluePrism or Automation Anywhere, etc. [13].

1) akaBot solution

akaBot is one of the RPA platforms to promote automation of simple operations, change the way of working and improve user satisfaction. The akaBot solution consists of several main products such as:

- **akaBot Studio:** is an automation process design tool that supports automation of many tasks and applications (such as automating web access, sending email, handling data files, opening and interacting on many applications installed on the desktop, etc.).
- **akaBot Agent:** is a tool that helps end users to simply enable Software Robots to work easily on their devices.
- **akaBot Center:** is a tool to set up a plan to operate the Software Robots according to the desired schedule.

2) Application of akaBot solution in process automation

The akaBot solution allows the development department to define the automation tasks that need to be performed in the process. Fig. 3. shows the overall design of the process of automatically sending survey comments to students through the akaBot solution. Some of the main work performed:

- **Init setting:** read the configuration parameters related to the system (username/password of the operator, the path of the web-based system containing the data to be accessed, etc.).
- **Get transaction data:** proceed to access the web-based system to get transaction data (which survey to take, email list, sending time, survey title, survey content, students who have not taken the survey need to resubmit the survey, etc.).
- **Process business:** conduct surveys to students.

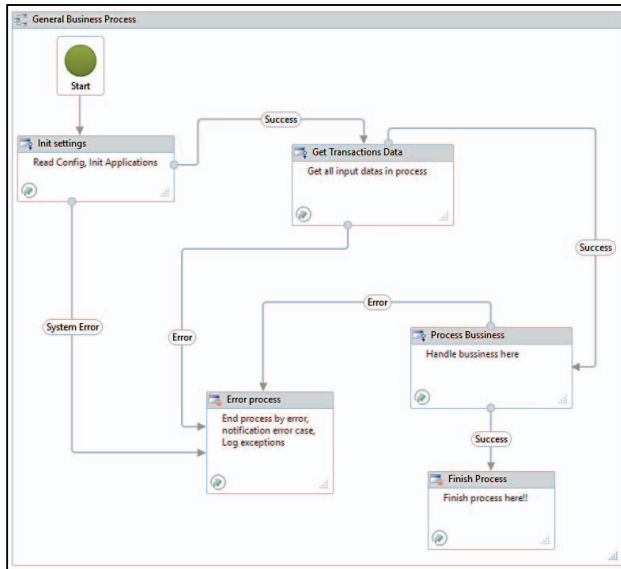


Fig. 3. Process of sending surveys to students based on akaBot solution

D. Performance Evaluation

Through the proposed implementation (1) development of web-based system development, initialization, feedback synthesis and data visualization and (2) RPA platform application (specifically akaBot) in automatic content submission the survey content to students helped to solve the first question posed by the study. The proposed implementation results also show that the application of the RPA platform is effective compared to manual implementation.

The research team has experimented with the project “Survey on satisfaction of incoming student about enrollment” of a university in Vietnam with a scale of 5,000 students. Survey content is updated on the web-based system, the sending of survey content to students is done automatically based on akaBot solution, survey results are updated and visualized in real time on the web-based system.

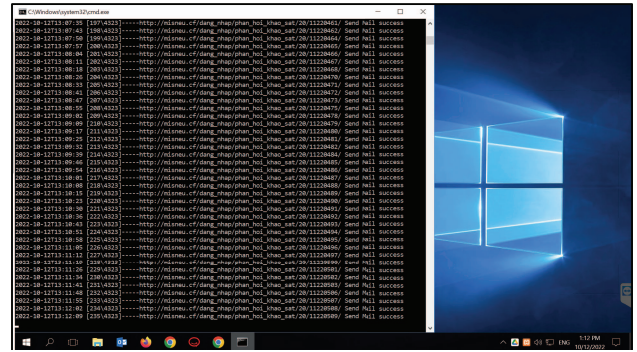


Fig. 4. Software robot automatically send surveys to students based on akaBot solution

Table 4 compares the results of sending surveys to students through automatic and manual ways. The results show that sending the survey automatically to students only takes an average of 7 seconds/student, 5 times more optimal than doing it manually. Besides, officers in charge feel more satisfied with their work compared to doing it manually.

TABLE IV. PERFORMANCE COMPARISON WHEN SENDING SURVEYS TO STUDENTS

	Sending surveys to students	
	Automated execution based on akaBot solution	Perform manually
Number of times performed	5,000 students	50 students (the research team performed 50 times the activity of sending manual survey email notifications to students to calculate the average time taken)
Average execution time	7 seconds / student	37 seconds / student
Total execution time (calculated with the volume done for 5000 students)	35,000 seconds	185,000 seconds
Number of errors generated	0 time	2 time (copy wrong data to send)

	Sending surveys to students	
	<i>Automated execution based on akaBot solution</i>	<i>Perform manually</i>
The level of readiness when doing the job	- Ready	- Hesitate (<i>large, boring workload</i>)
Level of job satisfaction	- Satisfied	- Unsatisfied
Advantages	- Survey email content is not moved to SPAM if sent in bulk. - Activate only 1 time. - Uniform data, updated in real time, no need to aggregate to build reports, quickly make decisions.	- Simple implementation, no need to think about complex deployment solutions.
Disadvantages	- Software robot development needs a team of qualified professionals, knowledgeable about RPA technology.	- Long implementation time, great effort. - Feeling bored from having to do the same thing over and over again. - Discrete data, it takes a lot of effort to aggregate to build a report, slow decision-making time.

However, the development of software robots will need a team knowledgeable about RPA technology and high determination of leadership. Besides, if combined with an integrated web-based (*configuring, aggregating and visualizing student survey data*) will bring great benefits in supporting universities to monitor student emotions in real time and quickly make decisions related to student policy.

IV. CONCLUSIONS

This research has developed a web-based system to create, synthesize and visualize data and a software robot to send survey data to students. The two systems above have been successfully integrated to help the university's leaders, staff, and lecturers track changes in student satisfaction. As a result, universities can quickly adjust appropriate policies, increase understanding, enhance cohesion and improve satisfaction with students. Through experimental activities to develop RPA solutions, it can be affirmed that RPA brings significant effects to optimize many practical problems in general and the field of education in particular. The immediate visible effects are the ability to automate tasks, reduce repetitive work, limit boredom and increase accuracy. Besides, the integration of RPA solutions with other systems in the organization is not too difficult - this is a significant advantage compared to implementing projects to add/upgrade automation features on other available systems (*deployment speed, cost, flexibility*).

However, the development of RPA projects requires a team knowledgeable about RPA technology as well as the determination from the university's leadership to boldly apply solutions in practice. In addition, the use of RPA platforms after the trial period also requires a certain amount of money in terms of copyright, which will also cause obstacles in implementation. If companies developing RPA solutions have specific support policies in the education sector (*like Google, Microsoft, etc.*), they can strongly promote the implementation of practical projects related to RPA technology.

V. FURTHER RESEARCH

The results obtained in the automation solution above can add the ability to automatically read, analyze, and classify content (*importance, positive/negative, unit in charge, ...*) from email feedback of students (*if students actively send more information*) and automatically transferred to the unit in charge for quick feedback to students. In addition, the dataset collected from this problem can be extracted to provide input data to help related units perform other independent data analysis activities. From there, it helps the university obtain a lot of useful information to improve the efficiency of management, training, providing services to students, etc.

REFERENCES

- [1] Training, V.M.o.E.a. *Statistics of Higher Education for the Academic year 2019-2020*. Available from: <https://moet.gov.vn/thong-ke/Pages/thong-ko-giao-duc-dai-hoc.aspx?ItemID=7389>.
- [2] Cooper, L.A., et al., *Robotic Process Automation in Public Accounting*. Accounting Horizons, 2019. **33**(4): p. 15-35.
- [3] Aguirre, S. and A. Rodriguez. *Automation of a business process using robotic process automation (RPA): A case study*. in *Workshop on engineering applications*. 2017. Springer.
- [4] Willcocks, L.P., I. Oshri, and J. Kotlarsky, *Dynamic Innovation in Outsourcing: Theories, Cases and Practices*. 2018: Springer.
- [5] Urbach, N., et al., *The impact of digitalization on the IT department*. Business & information systems engineering, 2019. **61**(1): p. 123-131.
- [6] Rolland, K.H., L. Mathiassen, and A. Rai, *Managing digital platforms in user organizations: The interactions between digital options and digital debt*. Information Systems Research, 2018. **29**(2): p. 419-443.
- [7] Sambamurthy, V., A. Bharadwaj, and V. Grover, *Shaping agility through digital options: Reconceptualizing the role of information technology in contemporary firms*. MIS quarterly, 2003: p. 237-263.
- [8] Willcocks, L.P., I. Oshri, and J. Kotlarsky, *Dynamic Innovation in Outsourcing - Theories, Cases and Practices*. Technology, Work and Globalization. 2018: Technology, Work and Globalization. XXX, 325.
- [9] Hevner, A. and S. Chatterjee, *Design science research in information systems*, in *Design research in information systems*. 2010, Springer. p. 9-22.
- [10] March, S.T. and G.F. Smith, *Design and natural science research on information technology*. Decision support systems, 1995. **15**(4): p. 251-266.
- [11] Peffers, K., et al., *A design science research methodology for information systems research*. Journal of management information systems, 2007. **24**(3): p. 45-77.
- [12] Kuechler, B. and V. Vaishnavi, *On theory development in design science research: anatomy of a research project*. European Journal of Information Systems, 2008. **17**(5): p. 489-504.
- [13] Map, C., *2019 RPA Chaos Map report*. 2019.